Ontology for Liberals

Hans Halvorson

June 29, 2017

"We not only believe the existence of atoms to be proved beyond a doubt, but also we even believe that we have an intimate knowledge of the constituents of the individual atoms."

An Alluring Idea

A well-defined **theory** must have a (primitive) **ontology**

i.e. an old-fashioned catalog of material objects

good	classical particle mechanics, Bohmian mechanics
bad	orthodox quantum mechanics, Copenhagen
ugly	many worlds QM

- The primitive ontology approach stresses the virtue of clear and unambiguous communication.
- But it runs the danger of **semantic fundamentalism**.

- **semantic fundamentalism:** drawing finer distinctions between theories than there are distinctions in the world.
- Examples
  - Demanding that a theory be written in German instead of English.
  - Demanding that we use polar coordinates instead of Cartesian coordindates.

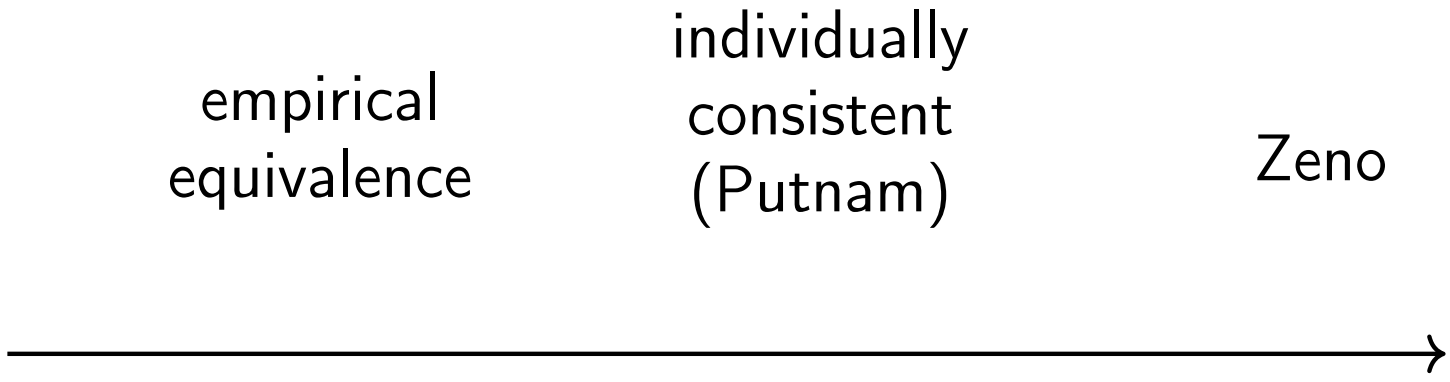
Main Questions

- When are **two** descriptions of reality **equivalent** to each other?
- What disputes are worth having?
- What kind of distinctions does the world care about?

left wing views

- **empirical equivalence**: If two theories make the same predictions, then they're equivalent.

left wing views

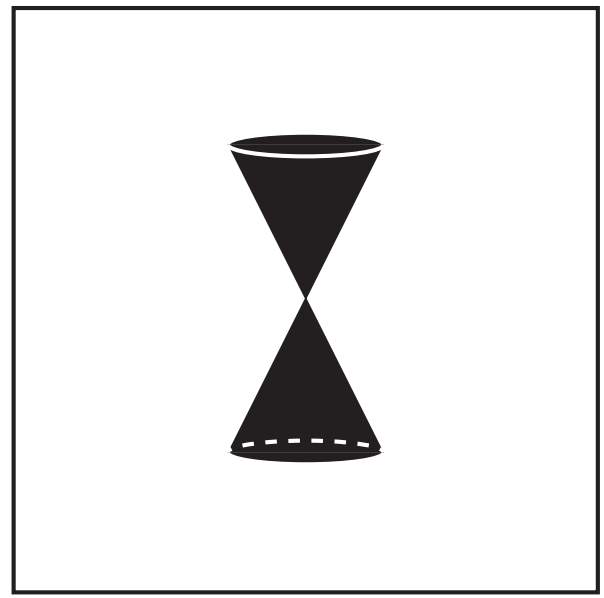


moving to the right

- primitive ontology vs wave-function monism
- wave mechanics vs matrix mechanics
- Hamiltonian mechanics vs Lagrangian mechanics

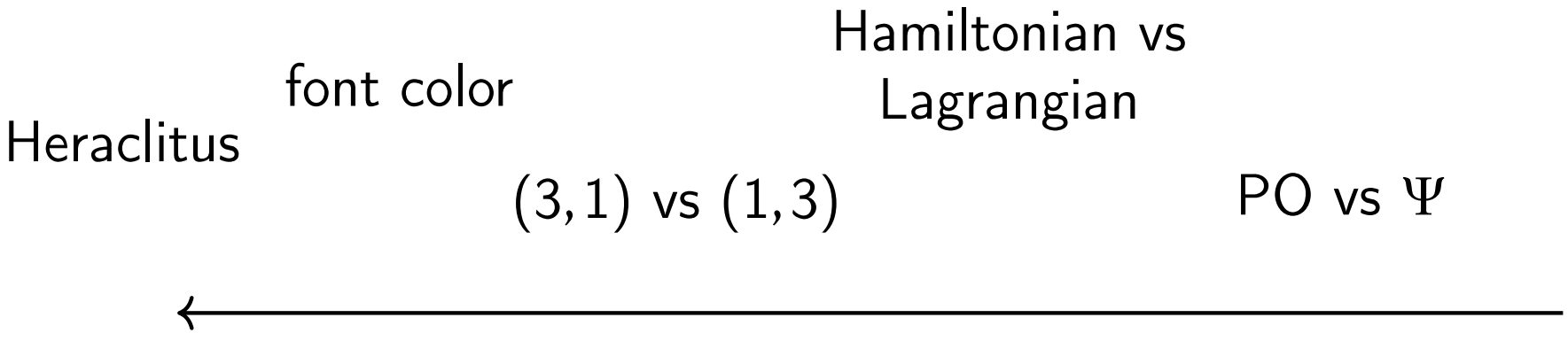


(1-3) general relativity



(3-1) general relativity

right wing views



Toward a clear account of theoretical equivalence

- Motivating idea: two theories are equivalent just in case they **say the same thing**.
- More precise: two theories are equivalent just in case they are **inter-translatable**.

A first-order language consists of:

- logical symbols $\forall, \exists, \wedge, \vee$, etc.
- variables $x_1, x_2, \ldots$
- a signature Σ consisting of specific relation and function symbols.

- A **reconstrual** of Σ into Σ' assigns to each relation symbol R of Σ a corresponding open-formula ϕ of Σ'.
- A reconstrual recursively generates a map from formulas of Σ to formulas of Σ'.
- This map is a **interpretation** of T into T' just in case $T \vdash \phi$ only if $T' \vdash \phi^*$.

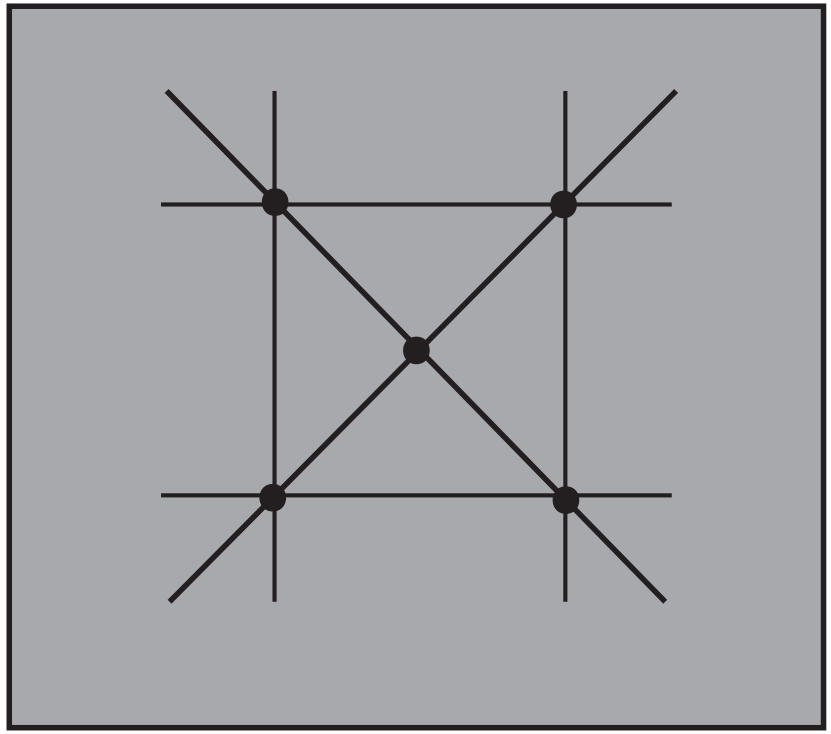
Two theories T and T' are **homotopy equivalent** just in case there are interpretations $F : T \to T'$ and $G : T' \to T$ such that:

$$\begin{aligned} T &\vdash \forall x(\phi(x) \leftrightarrow GF\phi(x)) \\ T' &\vdash \forall x(\psi(x) \leftrightarrow FG\psi(x)) \end{aligned}$$

Intuitive gloss

- Each model M of T corresponds to a unique model M' of T.
- M and M' are isomorphic — i.e. have the same number of objects, and they stand in the same relations.

Too conservative?

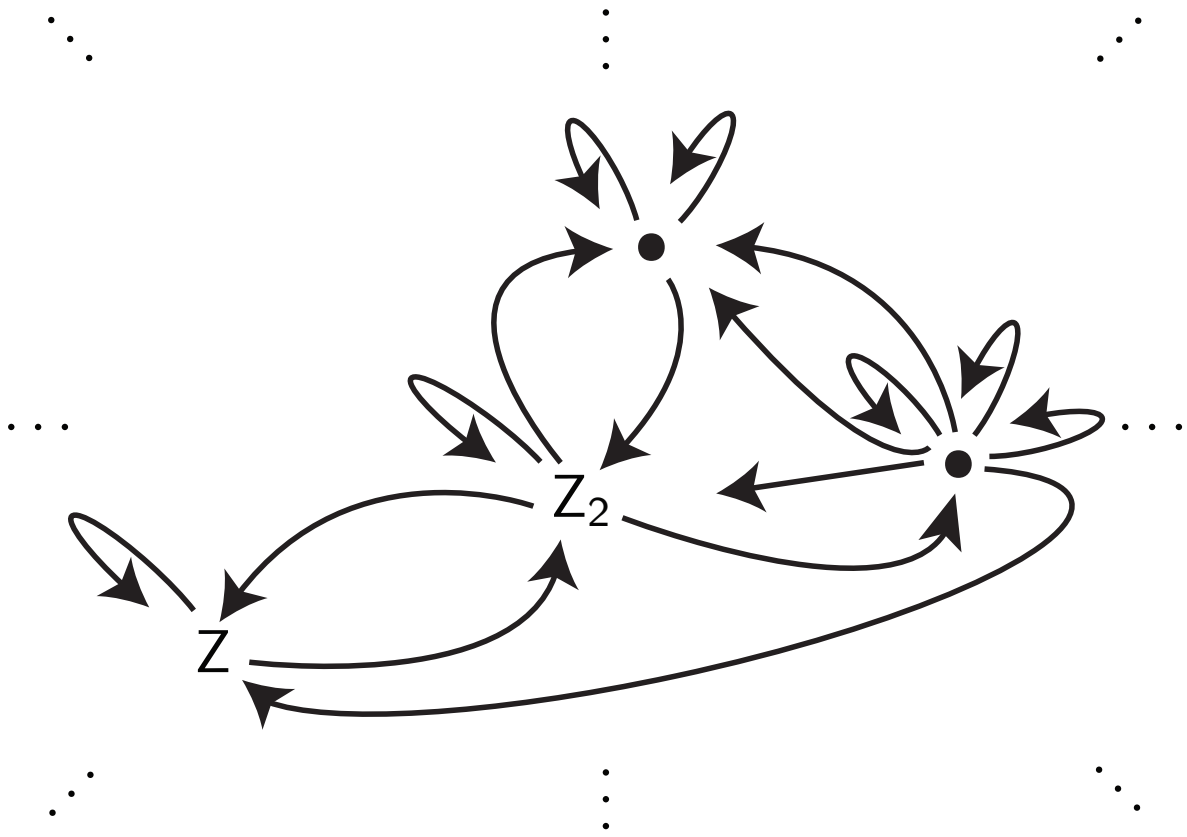


categorical equivalence

$\mathbf{Mod}(T)$ consists of two kinds of things:

objects models of T

arrows homomorphisms of models



categorical equivalence

objects of $\mathbf{C}$ $\longmapsto$ objects of $\mathbf{D}$

arrows of $\mathbf{C}$ $\longmapsto$ arrows of $\mathbf{D}$

- preserves arrow structure
- invertible, up to natural isomorphism

AZ equivalence

Generalized reconstrual: an n-ary relation symbol is mapped to an nm-ary formula

$$R(x_1, \ldots, x_n) \longmapsto \phi(\vec{y}_1, \ldots, \vec{y}_n)$$

AZ equivalence

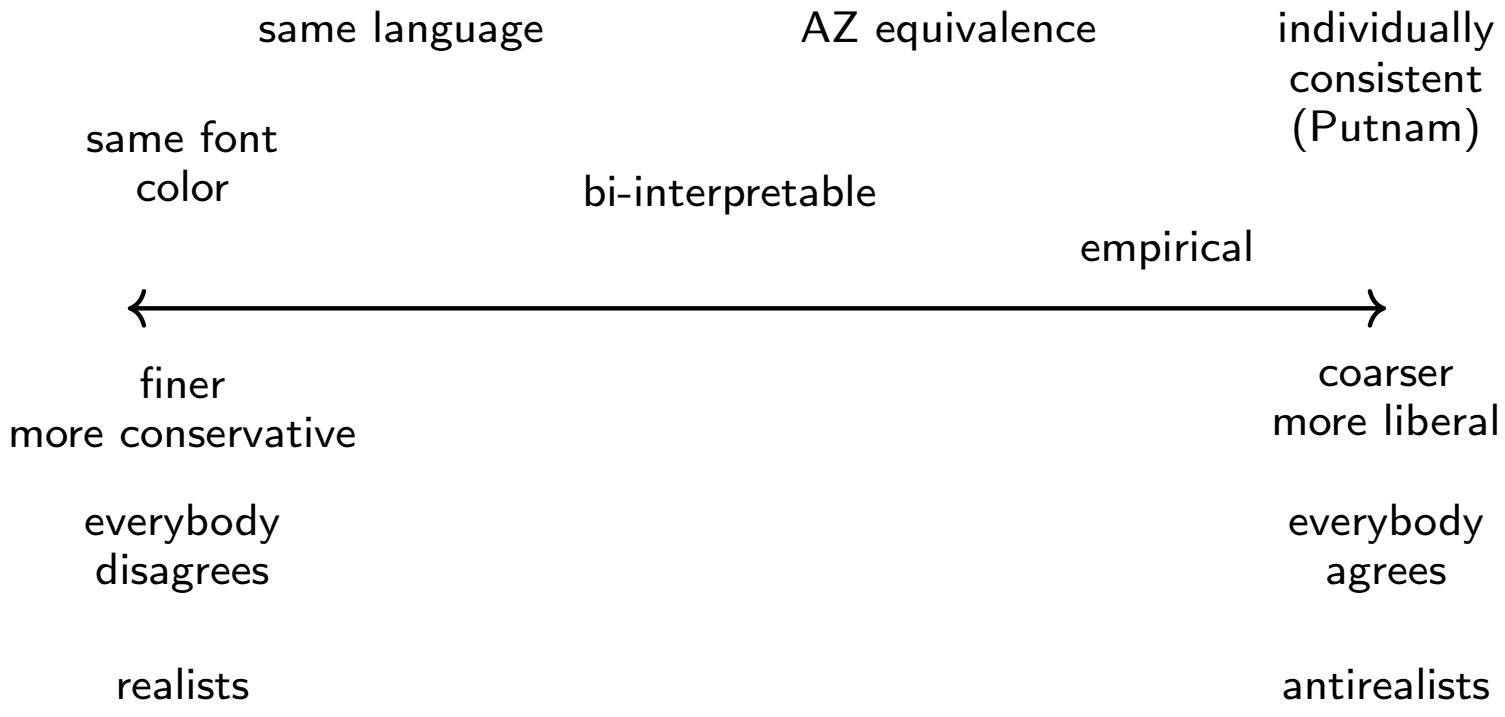
A homotopy equivalence between T and T' consists of a pair of reconstruals $F : T \to T'$, $G : T' \to T$, and connecting formulas η and ε such that:

$$\begin{aligned} T &\vdash \forall x(\eta(x,\vec{y}) \to (\phi(x) \wedge GF\phi(\vec{y}))) \\ T' &\vdash \forall x(\varepsilon(x,\vec{y}) \to (\psi(x) \wedge FG\psi(\vec{y}))) \end{aligned}$$

Intuitive gloss

- Each model M of T corresponds to a unique model M' of T'.
- But M and M' are **not** necessarily isomorphic.
- Every object of M' is a **logical construction** of objects of M, and vice versa.

the ideological spectrum



Take home points

- Our intuitive notion of theoretical equivalence calls for explication.
- When explication uses math, we may be guided by internal mathematical standards — which sometimes challenge naive intuitions. (Example: notions of infinity)

Take home points

- You can be completely clear about, and committed to the correctness of one ontology — while still conceding that there might be different, yet **equally correct**, ontologies.